

Ratul Das

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Personal Profile

Focused on numerical and data-driven modeling for fluid dynamics and heat transfer applications. Validated numerical models and optimized domain design for improved hydrothermal performance in undergraduate thesis research. Worked as a research assistant on data collection, model training, and quantization for deployment on microcontroller. Presented and published conference papers on CFD.

Education

Shahjalal University of Science and Technology(SUST), Sylhet, Bangladesh(*Passing year 2024*)
BSc(Engineering) in Mechanical Engineering, School of Applied Science and Engineering

- CGPA: 3.50/4.00
- **Coursework:** Applied Engineering Mathematics, Fluid Mechanics, Heat Transfer, Solid Mechanics, Materials, Statics and Dynamics, Numerical Analysis etc.

Research Experience

Undergraduate Thesis: Numerical Investigation of the Thermal and Hydraulic Performance of Plain Fin Compact Heat Exchangers Augmented with Constricted Flat Tubes.

Dept. of Mechanical Engineering, SUST

Supervisor: Mostafa Rafid, Lecturer, Mechanical Engineering, SUST.

- Validated a numerical model of a heat exchanger by comparing simulation results with published numerical and experimental data.
- Redesigned and optimized tube geometry, resulting in enhanced hydrothermal performance.

Research Project: Development of an Autonomous Robot for Floating Solid Waste Detection and Collection from Rivers Using Machine Learning Algorithms.

SUST Research Center

Principal Investigator: Md. Mahmud-Or-Rashid, Assistant Professor, Mechanical Engineering, SUST.

Co-investigator: Md. Syamul Bashar, Lecturer, Mechanical Engineering, SUST.

- Collected and preprocessed image data of water surface waste, and designed and built a double-hull catamaran for waste collection.
- Trained, optimized, and quantized an SSD MobileNet Lite object detection model for efficient deployment on a Raspberry Pi.

Publications

1. Tripta Sarker, Ratul Das, Sayed Tanvir Ahmed, and Mostafa Rafid. "Numerical investigation of the hydraulic and thermal performance of plain fin compact heat exchangers with modified flat tubes." In *American Institute of Physics Conference Series*, vol. 3307, no. 1, p. 020013. 2025. DOI: [10.1063/5.0262425](https://doi.org/10.1063/5.0262425) [🔗](#)
2. Sayed Tanvir Ahmed, Ratul Das, Tripta Sarker, and Mahadi Hasan Shanto. "Aerodynamic effects of leading-edge flap angle on NACA 4412 airfoil performance at low reynolds numbers: A CFD investigation." In *American Institute of Physics Conference Series*, vol. 3307, no.1, p. 020010. 2025. DOI:[10.1063/5.0262220](https://doi.org/10.1063/5.0262220) [🔗](#)
3. Sayed Tanvir Ahmed, Tripta Sarker, and Ratul Das. "Thermohydraulic performance optimization of solar air heater via tailored inverted L-shaped rib: A CFD investigation." In *American Institute of Physics Conference Series*, vol. 3307, no. 1, p. 030005. 2025. DOI: [10.1063/5.0262221](https://doi.org/10.1063/5.0262221) [🔗](#)

Test Scores

Graduate Record Examination (GRE): 304 (Quant:161, Verbal: 143)

Projects

Stepwise Finite Difference Implementation of Convection–Diffusion and Navier–Stokes. [link](#) [↗](#)

- Developed individual solvers for 1D and 2D convection and diffusion problems, progressively extending to 2D incompressible Navier–Stokes equations using Python. Applied the solvers to simulate [lid-driven cavity](#) and [channel flow](#) at low Reynolds numbers, using the projection method, with visualizations to illustrate flow dynamics.

Nonlinear 6-DOF RCAM Flight Dynamics Mathematical Model. [link](#) [↗](#)

- Implemented the mathematical equations of a nonlinear 6-DOF rigid-body aircraft model (RCAM) in MATLAB functions and integrated them into Simulink to run time-domain simulations with control inputs and dynamic state outputs.

Derivative Matrix Operator for 2D Laplace Equation via Kronecker Product. [link](#) [↗](#)

- Built a sparse matrix-based derivative operator using Kronecker product formulation and incorporated Dirichlet boundary conditions for the finite difference method to solve the 2D Laplace equation.

Mass-Spring-Damper System Simulation using Physics-Informed Neural Network (PINN). [link](#) [↗](#)

- Explored how physics-informed neural networks (PINNs) can predict the time evolution of a damped mass-spring system. Studied how different network structures affect training behavior and solution accuracy.

Skills

Programming Languages: C, C++, Python, Julia, Matlab, Octave.

Software: Ansys Fluent, Ansys ICEM, OpenFOAM, ParaView, Tecplot

3D Modeling: SolidWorks, AutoCAD, FreeCAD

Microcontroller: Raspberry Pi, Arudino

Hard Skill: Drill Press, Arc Welding, Cut-off saw, Lathe Machine

Others: Latex, Word, Excel, PowerPoint, Mathematica

Additional Courses

Computers, Waves, Simulations: A Practical Introduction to Numerical Methods using Python. [link](#) [↗](#)

- Learned finite difference, spectral, and element methods for PDEs, implemented mathematical models in Python, and performed benchmarking, convergence tests, and visualizations for accuracy.

Deep Learning Specialization. [link](#) [↗](#)

- Built and trained deep neural networks, applied TensorFlow for optimization, and developed CNNs for image detection, recognition, and style transfer.

Industrial Experience

Industrial Trainee

United Energy Limited, Sylhet

Industrial Trainee

Shahjalal Fertilizer Company Ltd. (SFCL), Sylhet

Teaching Experience

Private Tutor

SSC(O-Level), HSC(A-Level), and University Entrance Exams

- Provide one-on-one and group instruction in physics and mathematics, focusing on conceptual understanding and exam problem-solving skills. Successfully guided students in preparing for and passing competitive engineering university entrance exams.

Volunteering

Organising Secretary
Shikorh

In-Charge, Eng. Design
Kaizen SUST

Illumination Coordinator
Shikorh